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ChE 311 Quiz 3
Spring 2019

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Closed book and closed notes
5 minutes, 10 points

- ✓ 1. The flow defined by the following velocity field is incompressible flow.

$$\vec{v} = (t - x - y + z)\vec{\delta}_x + (2t + x + 2y + z)\vec{\delta}_y + (3t - x - y + z)\vec{\delta}_z$$

A. True

B. False

Answer: B

$$\begin{aligned}\frac{D\rho}{Dt} &= -\rho \nabla \cdot \vec{v} \\ &= -\rho \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \cdot (t - x - y + z, 2t + x + 2y + z, 3t - x - y + z) \\ &= -\rho (-1 + 2 + 1) = 0\end{aligned}$$

- X 2. The isothermal flow defined by the following velocity field in a cylindrical coordinate satisfies the continuity equation. The fluid is with constant density.

$$\vec{v} = (-r)\vec{\delta}_r + (2t + r + z)\vec{\delta}_\theta + (2z)\vec{\delta}_z$$

A. True

B. False

Answer: B

$$\begin{aligned}\frac{D\rho}{Dt} &= \rho \nabla \cdot \vec{v} \\ &= \rho (-1 + 1 + 2) = 2\rho \neq 0\end{aligned}$$

- X 3. Please write down Navier-Stokes function in vector format.

$$\frac{D\vec{v}}{Dt} = -\nabla P + \rho \vec{g} + \nabla^2 \vec{v}$$

4. Under which condition the initial term, $\rho \vec{v} \cdot \nabla \vec{v}$, in the N-S function can vanish as a reasonable approximation of the easier calculation.

- ✓ (A) Low Reynolds number, creeping flow.
(B) High Reynolds number, inviscid flow.
(C) Unidirectional flow, rectilinear motion.
(D) Turbulent flow.

Answer: C

- ✓ 5. Estimate the order of the Reynolds number of a swimmer who swims a kilometer in 20 minutes. Choose the closest answer.

(A) 0.01

(B) 1

(C) 100

(D) 10,000

Answer: A

$\frac{\rho v}{\mu}$